

Note 02 · Reading a size distribution: RRS and Gaudin

Two two-parameter curves that describe a whole feed; the numbers of Exercises 3 and 4 and of Assignment 1 by hand

1 The question this note answers

Every exercise in this course starts with a feed described by two numbers and a name: *RRS with $d_{63.2}$ 100 mm and n 1*, *RRS with $d_{63.2}$ 15 mm and n 0.9*, *RRS parameters d' 150 mm, λ 1.26*. This note explains what those two numbers mean, how to turn them into the cumulative percent finer that MODSIM wants at each size class, and how to read d_{80} , d_{99} or “percent finer than 40 mm” from them without opening any program.

2 A toy example before any theory

Take a bucket of gravel and sieve it on four screens. Suppose 10 % of the mass passes 2 mm, 40 % passes 5 mm, 70 % passes 10 mm and 95 % passes 20 mm. Plot “percent finer” against size and you get an S-shaped curve. Two facts make it inconvenient: it needs many points to be described, and it looks different on every plant. Engineers therefore fit a function with two parameters to it: one parameter says *where* the curve sits on the size axis (a characteristic size), the other says *how steep* it is (a spread). Two numbers replace a table.

Concept: The Rosin-Rammler-Sperling (RRS) distribution

$$P(d) = 1 - \exp \left[- \left(\frac{d}{d_{63.2}} \right)^n \right]$$

$P(d)$ is the mass fraction finer than size d . At $d = d_{63.2}$ the exponent is exactly 1 and $P = 1 - e^{-1} = 0.632$: hence the name, it is the size with 63.2 % finer. The sheets also call it d' . The exponent n (sometimes written λ) is the slope: a large n means a narrow distribution, a small n a wide one. With $n = 1$ the curve is a plain exponential.

Inverse: the size at which a fraction P passes is

$$d(P) = d_{63.2} \left[-\ln(1 - P) \right]^{1/n}.$$

That is how d_{80} ($P = 0.8$, so $-\ln 0.2 = 1.609$) and d_{99} ($-\ln 0.01 = 4.605$) are read.

Concept: The Gaudin (Gates-Gaudin-Schuhmann) distribution

$$P(d) = \left(\frac{d}{d_{100}} \right)^n$$

Here d_{100} is the size at which everything passes (the top size parameter) and n again the slope. It plots as a straight line on log-log paper, which is why the lecture calls the log-log diagram the Gaudin diagram, just as the RRS function plots straight on Rosin-Rammler paper. Both are “engineering art” in the lecture’s map of model philosophies: not physics, but very useful.

3 The three feeds of module 2, by hand

3.1 Exercise 3: $d_{63.2} = 100$ mm, $n = 1$

With $n = 1$: $P(d) = 1 - e^{-d/100}$.

Size (mm)	10	25	50	100	200	460
Cum % finer	9.5	22.1	39.3	63.2	86.5	99.0

$d_{80} = 100 \times 1.609 = 161$ mm and $d_{99} = 100 \times 4.605 = 460$ mm. Lesson for the size classes: the top class must reach about 500 mm, otherwise one percent of the feed has nowhere to go.

3.2 Exercise 4: $d_{63.2} = 15$ mm, $n = 0.9$

$P(d) = 1 - \exp[-(d/15)^{0.9}]$.

Size (mm)	2	5	10	15	25.5	82
Cum % finer	14.8	31.0	50.1	63.2	80	99

Half of the feed is already finer than 10 mm and a third finer than 5 mm. That single line is the design idea of Exercise 4: screen first at about 10 mm, send only the coarse half to the crusher. $d_{80} = 15 \times 1.609^{1/0.9} = 15 \times 1.697 = 25.5$ mm; $d_{99} = 15 \times 4.605^{1.111} = 82$ mm.

3.3 Assignment 1: $d' = 150$ mm, $n = 1.26$, top size 600 mm

$P(d) = 1 - \exp[-(d/150)^{1.26}]$.

Size (mm)	4	10	20	40	70	150	300	600
Cum % finer	1.0	3.2	7.6	17.2	31.8	63.2	90.9	99.7

Only 17 % of the 300 t/h feed is finer than 40 mm, so about 250 t/h have to pass through at least one crusher to make the 0-40 product. The products of that station are *made*, not screened out.

In one sentence

Two numbers, a characteristic size and a slope, replace the whole sieve table; $d_{63.2}$ is where 63.2 % passes, n says how wide the spread is, and $d(P) = d_{63.2}[-\ln(1 - P)]^{1/n}$ hands you any d_{80} or d_{99} in one line.

4 From the formula to MODSIM's feed form

MODSIM does not take $d_{63.2}$ and n ; it takes the cumulative percent finer at the upper size of each size class you defined. The auxiliary workbook **RRS.xls** (folder *auxiliary routines* in the material) does the evaluation for any pair of parameters and any list of sizes; **Gaudin.xls** does the same for the Gaudin form, and **Lin_Log.xls** converts between the plotting axes. Paste the column, do not type powers of e by hand.

The size classes themselves are a root-two series by tradition: each class boundary is $\sqrt{2}$ times the next (600, 424, 300, 212, 150, ...). Twenty classes from 600 mm down reach about 0.6 mm; that is what “20 fractions” in Assignment 1 means.

Watch out / typical mistake

- Interpolating on a linear size axis between table points gives wrong percentages for coarse feeds. Exercise 1 asks for 20 mm, which is not in its table: interpolate on

$\ln(d)$ (15 to 30 mm gives 42.3 %, not the 43.4 % a linear interpolation gives).

- $d_{63.2}$ is not d_{50} and not d_{80} . For Exercise 4 they are 15, 10 and 25.5 mm respectively.
- A top size class below the feed's d_{99} silently piles the coarse tail into the top class. This is exactly what happened in Exercise 1.mpd with a 10 mm top size against a 600 mm feed.
- The new GUI keeps sizes in metres. 150 mm is 0.150.

Try it yourself (keyboard in hand)

1. Open `RRS.xls`, enter $d_{63.2} = 15$ mm and $n = 0.9$, and a root-two list of sizes from 100 mm down to 0.05 mm. Check that the workbook gives 50 % at 10 mm and 63.2 % at 15 mm.
2. Change n to 1.5 and watch the curve narrow: the 5 mm point drops and the 25 mm point rises. That is what “slope” means.
3. For Assignment 1, compute by hand the tonnage coarser than 70 mm at 300 t/h ($1 - 0.318 = 0.682$, so 205 t/h) and compare it with what MODSIM shows in the feed fly-out once the feed is entered.

Glossary: in plain words

Cumulative percent finer the mass fraction of the material smaller than a given size; the y-axis of every size distribution plot in this course.

d_{80} , d_{99} the sizes at which 80 % and 99 % of the mass is finer; d_{80} is the size Bond's energy law uses.

Characteristic size the parameter that places a distribution on the size axis: $d_{63.2}$ for RRS, d_{100} for Gaudin.

Slope n the parameter that sets the spread; large n , narrow distribution.

Root-two series size classes whose boundaries differ by a factor $\sqrt{2}$; the standard sieve series and MODSIM's default.